THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 13: **Statistics** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 15	$\frac{10+20}{2}$
2	**(a)** $l + \frac{f_1-f_0}{2f_1-f_0-f_2} \times h$	Standard formula
3	**(a)** 26	$3(22) - 2(20)$
4	**(b)** 20–30	Highest frequency 12
5	**(b)** Incorrect — use $l + \frac{\frac{n}{2}-cf}{f} \times h$	Median needs the formula
6	**(a)** 2	$\frac{45-25}{10}$
7	**(a)** Student A is correct	B described the modal class
8	**(b)** 5.5	$\frac{55}{10}$
9	**(a)** 40	$\frac{1000}{25}$
10	**(c)** mode	Most frequent value

SECTION — B (Very Short Answer) — Answers 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\Sigma f = 20, \Sigma fx = 30 + 100 + 270 + 120 = 520 \Rightarrow$ mean $= 26$	1+1
12	Modal class 40–50: $40 + \frac{28-12}{56-12-20} \times 10 = 40 + \frac{16}{24} \times 10 = 46.67$	1+1
13	$n = 80, \frac{n}{2} = 40$; cf: 3, 12, 27, 57 ⇒ median class 30–40; modal class 30–40	1+1
14	$\frac{285+15p}{18+p} = 15.5 \Rightarrow 0.5p = 6 \Rightarrow p = 12$	1+1

SECTION — C (Short Answer) — Answers 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$a = 150$; $d = -40, -20, 0, 20, 40$; $\Sigma fd = -480 - 280 + 0 + 120 + 400 = -240$; mean $= 150 - \frac{240}{50} =$ ₹145.20	1+1+1
16	$n = 50$; cf 6, 16, 30 ⇒ class 20–30: $20 + \frac{25-16}{14} \times 10 = 26.43$	1+1+1
17	Modal class 35–45: $35 + \frac{23-21}{46-21-14} \times 10 = 35 + \frac{20}{11} = 36.82$ years	1+1+1
	OR: $x + y = 15$; $28.5 = 20 + \frac{30-(5+x)}{20} \times 10 \Rightarrow x = 8,\ y = 7$	1+1+1
18	$\Sigma f = 50, \Sigma fx = 100 + 320 + 720 + 1600 + 300 + 240 = 3280 \Rightarrow$ mean $= 65.6$. The student is wrong.	1+1+1
	OR: $a = 70,\ h = 10$; $\Sigma fu = -6 - 10 + 0 + 8 + 6 = -2$; mean $= 70 - \frac{20}{35} = 69.43\%$	1+1+1

SECTION — D (Long Answer) — Answers 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{1080}{40} = 27$ (ii) cf 4, 12, 22 ⇒ $20 + \frac{20-12}{10} \times 10 = 28$ (iii) $3(28) - 2(27) = 30$	2+2+1
	OR: $f_1 + f_2 = 52$; $3480 + 30f_1 + 70f_2 = 6000 \Rightarrow 3f_1 + 7f_2 = 252 \Rightarrow f_2 = 24,\ f_1 = 28$	2+2+1
20	(i) Modal 60–80: $60 + \frac{12-10}{24-10-6} \times 20 = 65$ (ii) $n = 50$, cf 6, 14, 24 ⇒ $60 + \frac{25-24}{12} \times 20 = 61.67$ (iii) No — he likely took the wrong modal class	2+2+1
	OR: Median: cf 2, 10, 22 ⇒ $150 + \frac{25-22}{20} \times 10 = 151.5$ cm. Mean: $a = 145, h = 10, \Sigma fu = 24 \Rightarrow 149.8$ cm	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 13: **Statistics** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 20–30	25 ± 5
2	**(a)** $l + \frac{\frac{n}{2} - cf}{f} \times h$	Standard formula
3	**(a)** 48	$\frac{24+2(60)}{3}$
4	**(a)** $x_i - a$	Deviation from assumed mean
5	**(b)** Incorrect — it can be greater, smaller or equal	Depends on the data
6	**(b)** 20–30	$\frac{n}{2} = 15$ first exceeded at 18
7	**(c)** Both get the same mean	Mean does not depend on the choice of a
8	**(a)** 54	$50 + 10 \times 0.4$
9	**(b)** the sum of frequencies up to and including that class	Definition
10	**(a)** 17	$\frac{90-22}{4}$

SECTION — B (Very Short Answer) — Answers 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\Sigma fx = 35 + 120 + 300 + 455 = 910$; mean $= \frac{910}{40} = 22.75$	1+1
12	Modal 40–60: $40 + \frac{18-6}{36-6-10} \times 20 = 52$	1+1
13	$n = 60$; cf 6, 15, 35 ⇒ median class 40–60; modal class 40–60	1+1
14	$\Sigma fx = 30 + 100 + 210 + 360 = 700$, $\Sigma f = 20$ ⇒ mean $= 35$. Claim is false (frequencies ignored).	1+1

SECTION — C (Short Answer) — Answers 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$a = 225,\ h = 50$; $\Sigma fu = -8 - 5 + 0 + 2 + 4 = -7$; mean $= 225 - \frac{350}{25} =$ ₹211	1+1+1
16	$\frac{n}{2} = 200$; cf 14, 70, 130, 216 ⇒ $3000 + \frac{70}{86} \times 500 = 3406.98$ hrs	1+1+1
17	Modal 60–80: $60 + \frac{61-52}{122-52-38} \times 20 = 60 + 5.625 = 65.625$	1+1+1
	OR: Modal 30–40: $30 + \frac{16-x}{32-x-12} \times 10 = 36 \Rightarrow \frac{16-x}{20-x} = 0.6 \Rightarrow x = 10$	1+1+1
18	Classes 140–145: 7, 145–150: 18, 150–155: 11, … $\frac{n}{2} = 25.5$ ⇒ class 145–150: $145 + \frac{25.5-11}{18} \times 5 = 149.03$ cm	1+1+1
	OR: $p + q = 20$; $2060 + 30p + 70q = 3140 \Rightarrow 3p + 7q = 108$ ⇒ $q = 12,\ p = 8$	1+1+1

SECTION — D (Long Answer) — Answers 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) cf 5, 15, 33 ⇒ $20 + \frac{25-15}{18} \times 10 = 25.56$ (ii) $a = 25$, $\Sigma fu = -10 - 10 + 0 + 12 + 10 = 2$ ⇒ 25.4 (iii) He probably took the class 30–40 (after the median class) or used n instead of $\frac{n}{2}$	2+2+1
	OR: Median: cf 2, 10, 22, 46 ⇒ $70 + \frac{50-46}{38} \times 5 = 70.53$. Mode: $70 + \frac{38-24}{76-24-16} \times 5 = 71.94$	3+2
20	(i) 800–1000 (ii) $800 + \frac{15-10}{30-10-12} \times 200 = 925$ (iii) $a = 900$, $\Sigma fd = -2800 - 2000 + 0 + 2400 + 2400 = 0$ ⇒ mean = ₹900	1+2+2
	OR: $x + y = 24$; median class 500–600: $525 = 500 + \frac{50-(36+x)}{20} \times 100 \Rightarrow x = 9,\ y = 15$	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.